

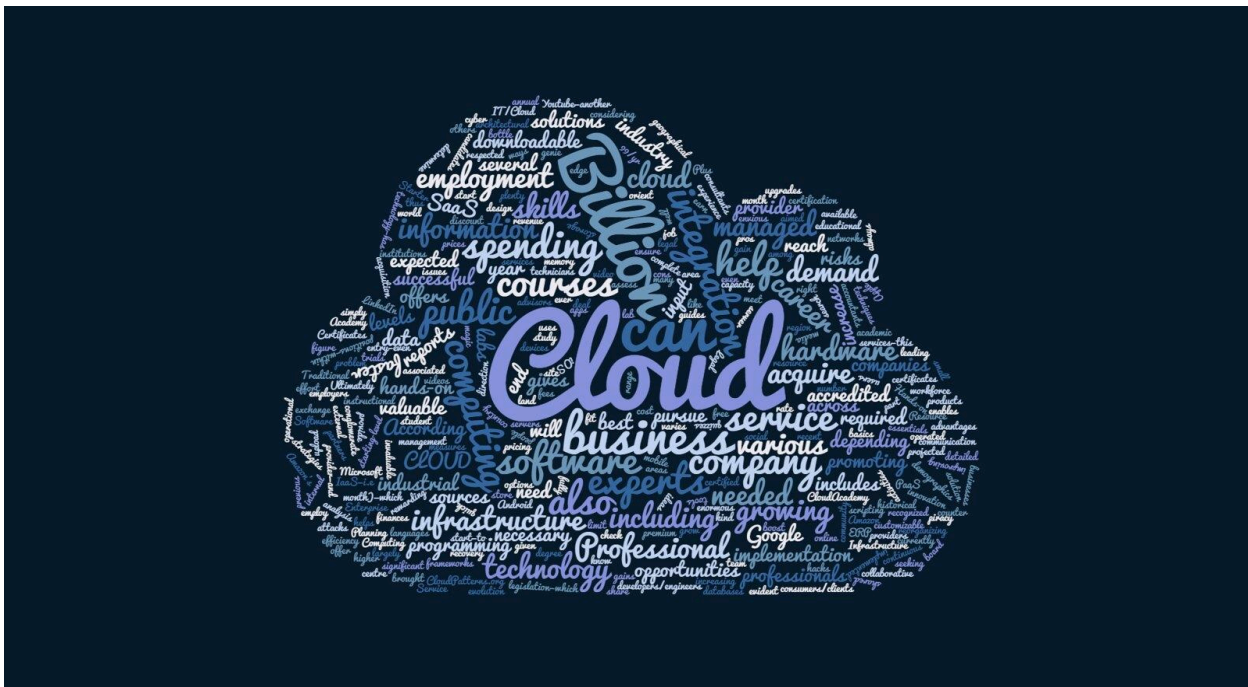
Cloud computing

What is cloud computing?

Cloud Computing means storing and accessing the data and programs on remote servers that are hosted on the internet instead of the computer's hard drive or local server. Cloud computing is also referred to as Internet-based computing, it is a technology where the resource is provided as a service through the Internet to the user. The data that is stored can be files, images, documents, or any other storable document.

The following are some of the Operations that can be performed with Cloud Computing

1. **Storage, backup, and recovery of data**
2. **Delivery of software on demand**
3. **Development of new applications and services**
4. **Streaming videos and audio**



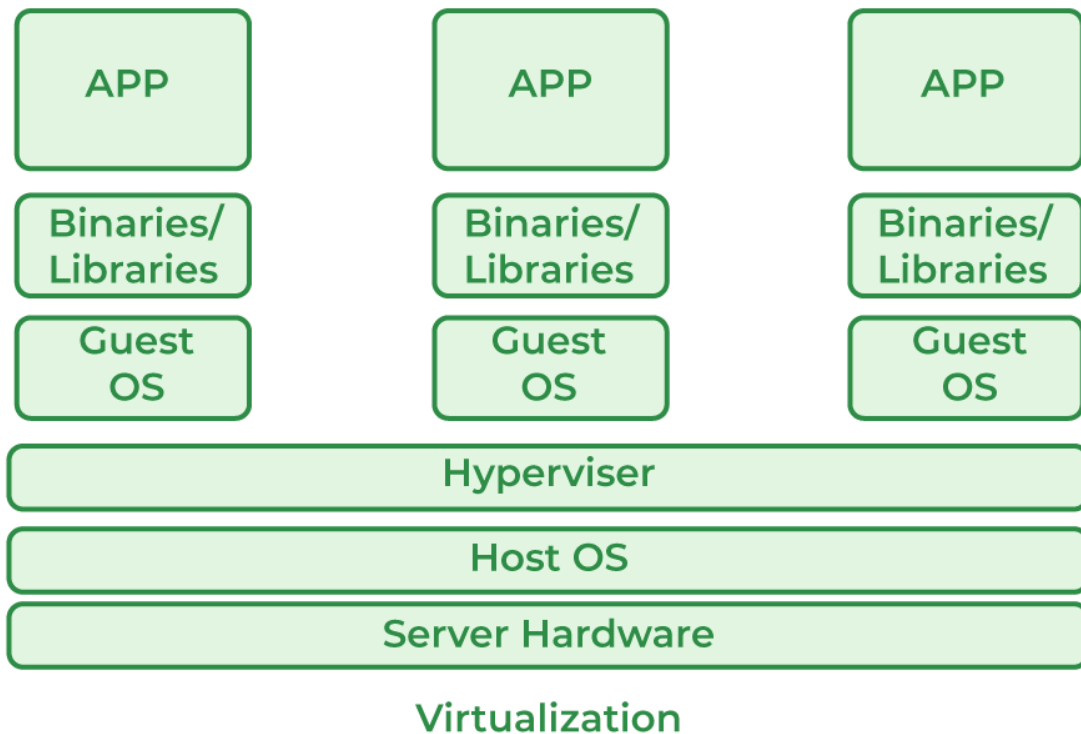
Key concept

1. Virtualization
2. History of cloud ?
3. Types of cloud?

4. Types of cloud service ?

What is virtualization ?

- ❖ Virtualization is a technique that creates virtual versions of computer hardware or resources, enabling multiple operating systems and applications to run on the same physical machine simultaneously. It enhances hardware utilization and flexibility, reducing costs, hardware requirements, and energy consumption. Cloud providers extensively use virtualization to share resources among multiple customers, assigning logical names to physical storage and providing pointers to those resources as needed. It's fundamental for Infrastructure-as-a-Service (IaaS) solutions in cloud computing, creating virtual environments for applications, storage, memory, and networking.



Types of virtualization

1. Application Virtualization: Application virtualization helps a user to have remote access to an application from a server. The server stores all personal information and other characteristics of the application but can still run on a local workstation through the internet. An example of this would be a user who needs to run two different versions of the same software. Technologies that use application virtualization are hosted applications and packaged applications.

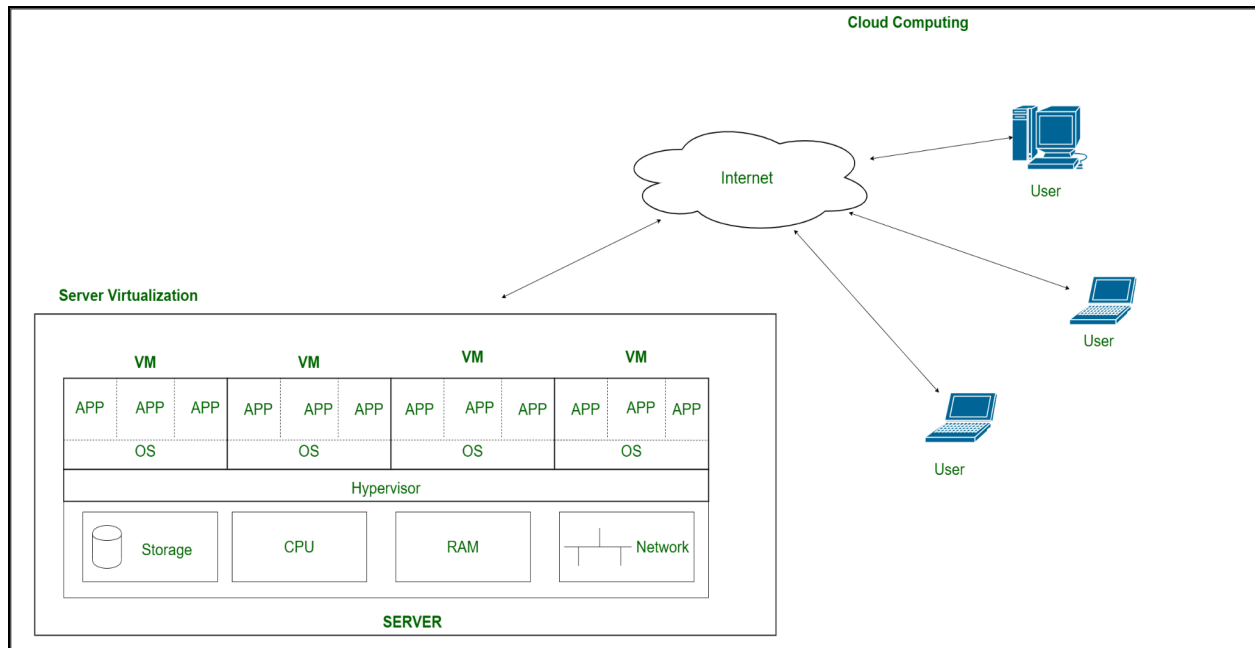
2. Network Virtualization: The ability to run multiple virtual networks with each having a separate control and data plan. It co-exists together on top of one physical network. It can be managed by individual parties that are potentially confidential to each other. Network virtualization provides a facility to create and provision virtual networks, logical switches, routers, firewalls, load balancers, Virtual Private Networks (VPN), and workload security within days or even weeks.

3. Desktop Virtualization: Desktop virtualization allows the users' OS to be remotely stored on a server in the data center. It allows the user to access their desktop virtually, from any location by a different machine. Users who want specific operating systems other than Windows Server will need to have a virtual desktop. The main benefits of desktop virtualization are user mobility, portability, and easy management of software installation, updates, and patches.

4. Storage Virtualization: Storage virtualization is an array of servers that are managed by a virtual storage system. The servers aren't aware of exactly where their data is stored and instead function more like worker bees in a hive. It makes managing storage from multiple sources be managed and utilized as a single repository. storage virtualization software maintains smooth operations, consistent performance, and a continuous suite of advanced functions despite changes, breaks down, and differences in the underlying equipment.

5. Server Virtualization: This is a kind of virtualization in which the masking of server resources takes place. Here, the central server (physical server) is divided into multiple different virtual servers by changing the identity number, and processors. So, each system can operate its operating systems in an isolated manner. Where each sub-server knows the identity of the central server. It causes an increase in performance and reduces the operating cost by the deployment of main server resources into a sub-server resource. It's beneficial in virtual migration, reducing energy consumption, reducing infrastructural costs, etc.

6. Data Virtualization: This is the kind of virtualization in which the data is collected from various sources and managed at a single place without knowing more about the technical information like how data is collected, stored & formatted then arranged that data logically so that its virtual view can be accessed by its interested people and stakeholders, and users through the various cloud services remotely. Many big giant companies are providing their services like Oracle, IBM, At scale, Cdata, etc.



Benefits of virtualization

1. **1. Protection From System Failures:**Virtualization in cloud computing provides protection against system failures by automatically backing up data across multiple devices in the virtualized cloud network. This ensures that data remains accessible from any device, minimizing the risk of data loss and preserving work progress even during system failures.
2. **Firewall and Security Support:**Virtualization, along with virtual firewalls, offers cost-effective cybersecurity solutions by restricting access to data. This approach provides premium data protection without additional expenses compared to traditional methods.
3. **Cost-Effective Strategies:**Virtualization in cloud computing reduces operational costs by eliminating the need for multiple physical servers, cutting down on maintenance and electricity expenses. It also saves server space for other crucial tasks, leading to efficient resource utilization and reduced waste
4. **Disaster Recovery Is Efficient and Easy:**In virtualization, data is always backed up on the cloud, simplifying recovery and enabling easy

duplication, unlike physical servers where data loss can occur with significant recovery time and effort.

- 5. Streamlined Processing and Operations** :Virtualization centralizes resources and management. Which makes it easier for the IT team to maintain the system in a more streamlined way. Instead of juggling individual devices, which can be complicated, they can manage their operations from a single source. Repair, software installation, patching and maintenance become much easy and less time-consuming. So it frees up your IT team to focus their energy elsewhere

Challenges in virtualization

While virtualization offers numerous benefits, there are also challenges to consider. These include **security concerns**, **performance issues**, **vendor lock-in**, and **complexity of management**. It is crucial for organizations to address these challenges effectively to fully leverage the power of virtualization.

Failure of virtualization

- 1. Resource distribution**

The way virtualization partitions systems can result in varied ways — some might function really well, and others might not provide users access to enough resources to meet their needs. Resource distribution problems often occur in the shift to virtualization and can be fixed by working on capacity planning with your service provider.

- 2. VM Sprawl**

VM sprawl, the unchecked growth of virtual machines in a virtual environment, as any virtualization admin knows, can cripple an otherwise healthy environment. It is problematic because its underlying cause often stays hidden until it manifests in resource shortages.

You should look at how virtual machines will be managed, who will be doing what, and what systems you're going to use. One of the optimal times to develop an overall management plan is when you're in a testing phase, before migration.

- 3. Backward compatibility**

Using legacy systems can cause problems with newer virtualized software programs. Compatibility issues can be time-consuming and difficult to solve. A good provider may be able to suggest upgrades and workarounds to ensure that everything functions the way they should.